

COURSE SLIDES · WSQ

IoT with Raspberry Pi

WSQ Course Code: TGS-2020505433

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

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COURSE ADMINISTRATION

Welcome & Housekeeping

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Digital Attendance (Mandatory)

1

MORNING (AM)
Scan on arrival

2

AFTERNOON (PM)
Scan after lunch

3

ASSESSMENT
Scan before you begin

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How Digital Attendance Works

1

Mandatory for WSQ funding

AM, PM and Assessment attendance must all be taken, on both training days.

2

The QR code

The trainer or administrator displays the QR code generated from the SSG portal.

3

How to submit

Scan the QR code with your mobile phone camera and submit your attendance.

4

75% minimum

At least 75% attendance is required to be eligible for assessment and funding.

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About the Trainer



Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

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YOUR TRAINER

About the Trainer



Dr. Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte Ltd

ROLE
Principal Trainer, Tertiary Infotech Academy Pte Ltd

EXPERTISE
Internet of Things, embedded systems and cloud-connected devices.

DELIVERS
WSQ courses on IoT, Raspberry Pi, microcontrollers, AI and data analytics.

FOUNDER
Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

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ICE-BREAKER

Let's Know Each Other

- Your name, organisation and role.
- Your experience with Raspberry Pi, electronics or programming — beginners are very welcome.
- One thing you would like to measure, monitor or control by the end of this course.

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HOUSEKEEPING

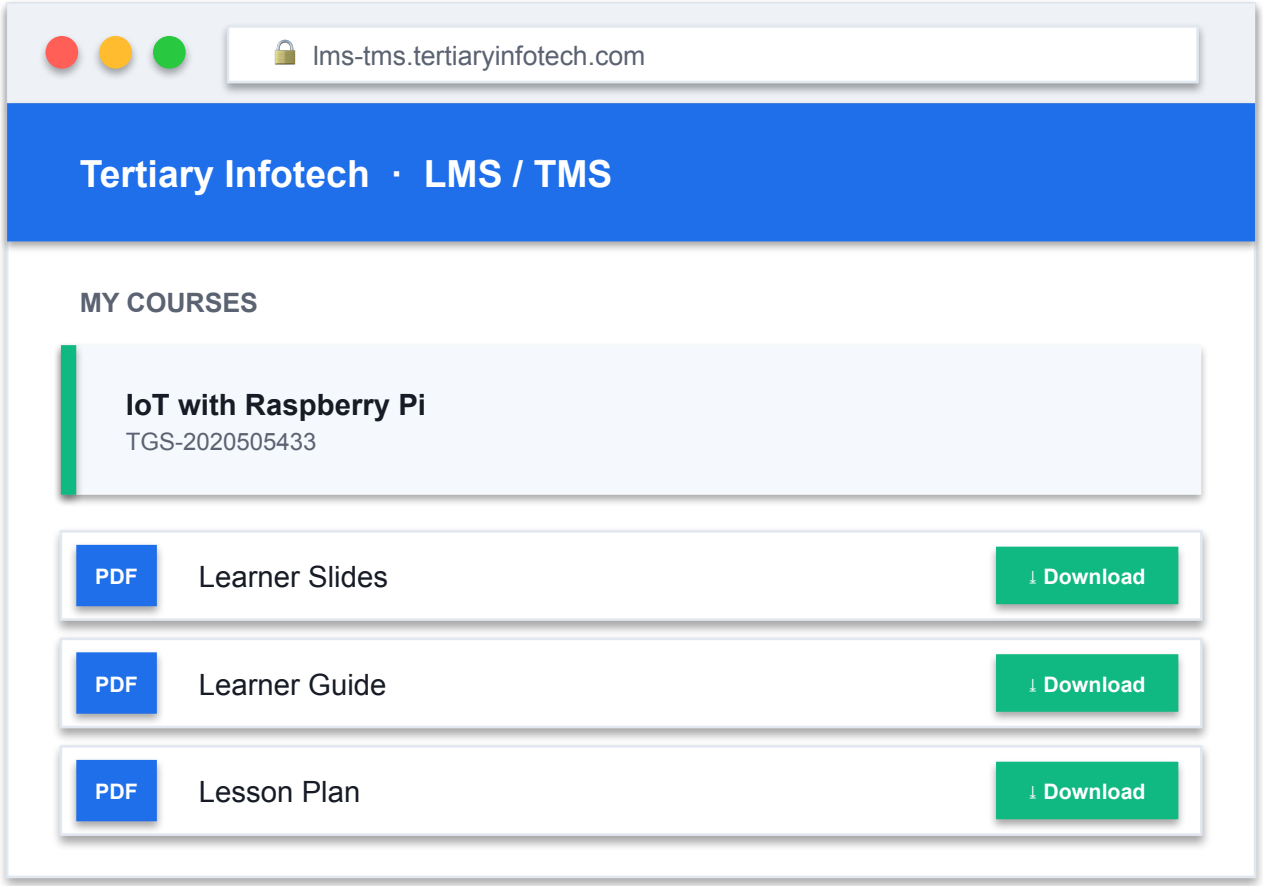
Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively — no question is too small.
3. Mutual respect: agree to disagree.
4. One conversation at a time.
5. Be punctual; return from breaks on time.
6. 75% attendance is required.

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COURSE PORTAL

Download Your Course Material



1. **Go to lms-tms.tertiaryinfotech.com**
   Open the portal in any browser.
2. **Log in with your registered email**
   The same address used to enrol.
3. **Open this course**
   IoT with Raspberry Pi.
4. **Download the material**
   Slides, Learner Guide and Lesson Plan.
5. **Keep them open in class**
   The assessment is open book.

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Skills Framework — TSC

1

TSC Title

Internet of Things Management

2

TSC Code

EVS-TEM-4002-1.1

3

Sector

Environmental Services

4

Proficiency

Level 4 — Manage the deployment of IoT solutions to monitor and control equipment and the environment, and to drive process improvement.

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TSC Abilities — Assessed by the Practical Performance

1. **A1**
   Analyse relevant data to recommend control actions

2. **A2**
   Identify areas for implementing IoT solutions for process improvement

3. **A3**
   Use relevant tool to analyse and record performance

4. **A4**
   Monitor the effectiveness of IoT solutions

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TSC Knowledge — Assessed by the Written Assessment

1

K1

IoT concept and technical knowledge of IoT implementation

2

K2

Connectivity in devices using sensors, smart devices and other technologies for data collection and communication

3

K3

Equipment automation

4

K4

Building and/or Environment automation

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TSC Knowledge — Assessed by the Written Assessment (cont.)

1. **K5**
   Advanced process control

2. **K6**
   Security and privacy applications for IoT

3. **K7**
   IoT guidelines and communication standards

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SCHEDULE · DAY 1 · 9:30am – 6:30pm

Lesson Plan — Day 1

Morning · 9:30am – 12:30pm

- Digital Attendance (AM) · Introductions
- Learning outcomes, outline & TSC briefing
- Topic 1: Overview of IoT (LO1)
- Discussion — Analyse an IoT Use Case
- Lunch Break — 1 hour

Afternoon · 1:30pm – 6:30pm

- Digital Attendance (PM)
- Topic 2: Collect & Post Data to Cloud (LO2)
- Lab 1: Register Your Raspberry Pi
- Lab 2: Collect & Post Environmental Data
- Day 1 recap and Q&A

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SCHEDULE · DAY 2 · 9:30am – 6:30pm

Lesson Plan — Day 2

Morning · 9:30am – 12:30pm

- Digital Attendance (AM) · Day 1 recap
- Topic 3: Data Analytics & Visualization (LO3)
- Lab 3: Analyse & Visualise Your IoT Data
- Discussion — Design Your Dashboard
- Lunch Break — 1 hour

Afternoon · 1:30pm – 6:30pm

- Digital Attendance (PM)
- Topic 4: Monitor Data & Trigger Control (LO4)
- Lab 4: Monitor Data and Trigger Control
- TRAQOM Survey · Assessment attendance
- Final Assessment — WA (60 min) + PP (60 min)

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WHAT YOU'LL ACHIEVE

Learning Outcomes

1

LO1 · Analyse IoT technologies

Analyse the uses and functions of IoT technologies. (A1, K1)

2

LO2 · Collect and post data

Collect environment data with Raspberry Pi and post it to cloud computing. (A2, K2, K3)

3

LO3 · Analyse and visualise

Use cloud computing to analyse and visualise IoT data. (A3, K4, K5)

4

LO4 · Monitor and control

Monitor data and trigger control on devices. (A4, K6, K7)

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FOUR TOPICS · TWO DAYS · FOUR HANDS-ON LABS

Course Outline

1

Topic 1 — Overview of IoT

Industry 4.0, what IoT is, sensors and actuators, wireless technologies, use cases.

2

Topic 2 — Collect & Post Data

Cloud platforms, Raspberry Pi as gateway, device tokens, HTTP REST and MQTT uplink.

3

Topic 3 — Analytics & Visualization

Dashboards, widgets, telemetry history, trends and alert rules.

4

Topic 4 — Monitor & Trigger Control

Downlink control, virtual pins, n8n automation, effectiveness and IoT security.

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Hands-On Labs on IoTFlow

Day 1 · Labs 1–2

- Register your Raspberry Pi
- Collect and post sensor data
- HTTP REST and MQTT uplink

Day 2 · Lab 3

- Build a live dashboard
- Analyse telemetry history
- Create alert rules

Day 2 · Lab 4

- Remote control over virtual pins
- n8n automation and AI
- Close the monitor-act loop

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Briefing for Assessment

1. Place phones and other materials under the table or on the floor.
2. No photos or recording of assessment scripts.
3. No discussion during the assessment.
4. Use a black or blue pen for hard-copy assessments.
5. No liquid paper or correction tape.
6. Scripts are collected when the time is up.

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ASSESSMENT · 120 MINUTES TOTAL · END OF DAY 2

Final Assessment

WA (Q&A) — 60 min

- Short-answer questions
- Assesses knowledge K1–K7
- Open book

Practical Performance — 60 min

- Build a working IoT solution
- Assesses abilities A1–A4
- Open book, done individually

What you may use

- The course slides
- Your Learner Guide
- Approved materials only
- An appeal process is available

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ON ASSESSMENT DAY

Assessment Flow

1. TRAQOM survey — scan the QR code on the LMS
2. Assessment digital attendance — scan the SSG QR
3. Sit the WA (Q&A) — 60 min, open book
4. Complete the Practical Performance — 60 min
5. Submit your answers on the LMS
6. Sign the Assessment Summary Record

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1

DAY 1

IoT Foundations and Collecting Data — From Sensor to Cloud

Topic 1: Overview of IoT · Topic 2: Collect and Post Environmental Data to Cloud

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01

TOPIC 01

Overview of Internet of Things (IoT)

Industry 4.0 · What IoT is · Sensors & actuators · Wireless technologies · Use cases

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WHY NOW

We are in the 4th Industrial Revolution.

Steam, electricity, computers — and now cyber-physical systems where physical assets report their own state in real time.

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Key Concepts — Overview of Internet of Things (IoT)

1

Industry 4.0

Steam, electricity, computers and now cyber-physical systems — assets report their own state in real time.

2

What IoT is

A network of physical objects embedded with sensors and connectivity that collect and exchange data.

3

The four building blocks

Sensors/devices capture data, connectivity moves it, cloud processes it, a UI acts on it.

4

Sensors and actuators

Sensors turn a physical phenomenon into a signal; actuators turn a decision back into physical action.

5

Wireless technologies

WiFi, BLE, Zigbee, LoRaWAN, NB-IoT and 5G trade range against power, bandwidth and cost.

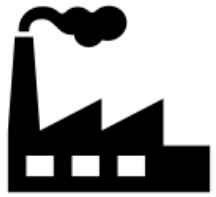
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Types of IoT

Consumer, commercial, industrial (IIoT) and infrastructure IoT each serve a different context.

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The Fourth Industrial Revolution



Industry 1.0
Mechanization, steam power, weaving loom

(18th century, 1760 - 1840)



Industry 2.0
Mass production, assembly line, electrical energy

(19th century, 1871 - 1914)



Industry 3.0
Automation, computers and electronics

(late 20th century, 1970 - present)



Industry 4.0
Cyber Physical Systems, internet of things, networks

(early 21st century, 2011 - present)

Industry 4.0 — cyber-physical systems, IoT, cloud computing and cognitive computing.

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FROM STEAM TO SENSORS

The Four Industrial Revolutions



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Technologies Driving Industry 4.0

1

Industrial IoT (IIoT)

Connected machines and sensors on the factory or plant floor.

2

Big data & analytics

Turning high-volume machine data into operational decisions.

3

Cloud computing

Storage, compute and availability without running your own servers.

4

Cybersecurity

Protecting devices, data and the physical processes they control.

5

Autonomous robots

Machines that sense their environment and act without an operator.

6

Digital twin

A live simulation of a physical asset, fed by its own telemetry.

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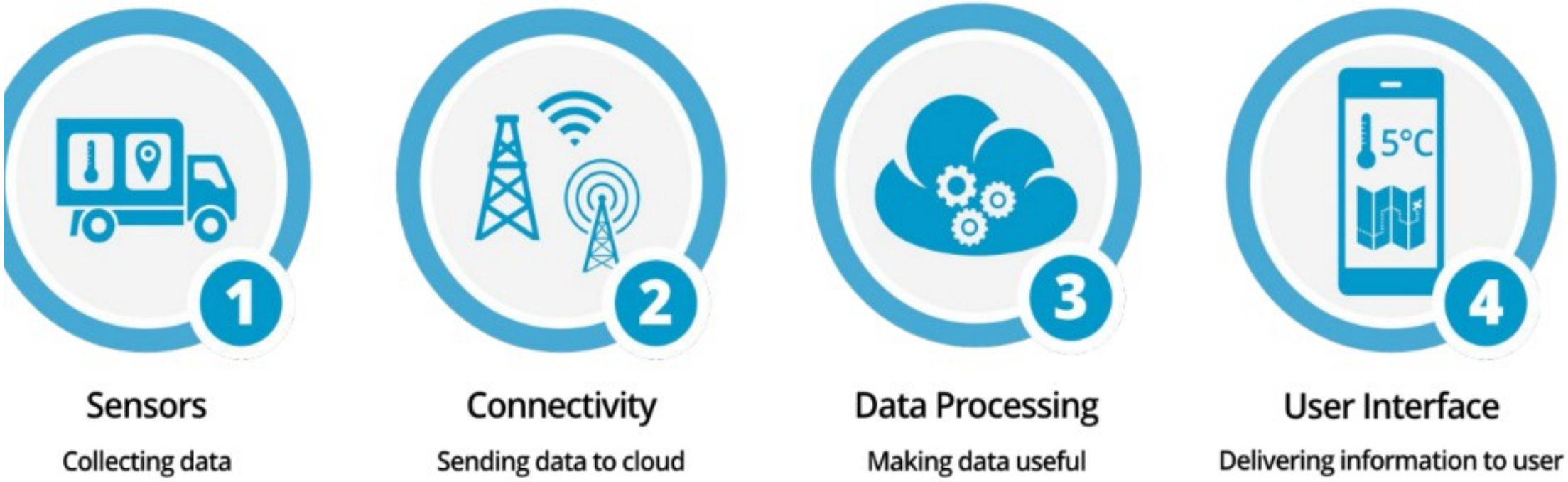
What Is the Internet of Things?



IoT connects everyday objects to the internet so they can send and receive data.

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The Components of an IoT System



A 'thing' is any object with embedded electronics that transfers data over a network.

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THE OPPORTUNITY

The Scale of IoT

$9T

Estimated IoT market value

50B+

Connected devices worldwide

4th

Industrial revolution underway

5G

The accelerator to mass adoption

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Driving Forces of IoT



Low-cost sensors and long-range wireless made IoT viable; 5G is driving mass adoption.

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How IoT Works — The Four Building Blocks

1

SENSORS & DEVICES capture data from the physical world

2

CONNECTIVITY carries it over WiFi, cellular, LoRa or MQTT

3

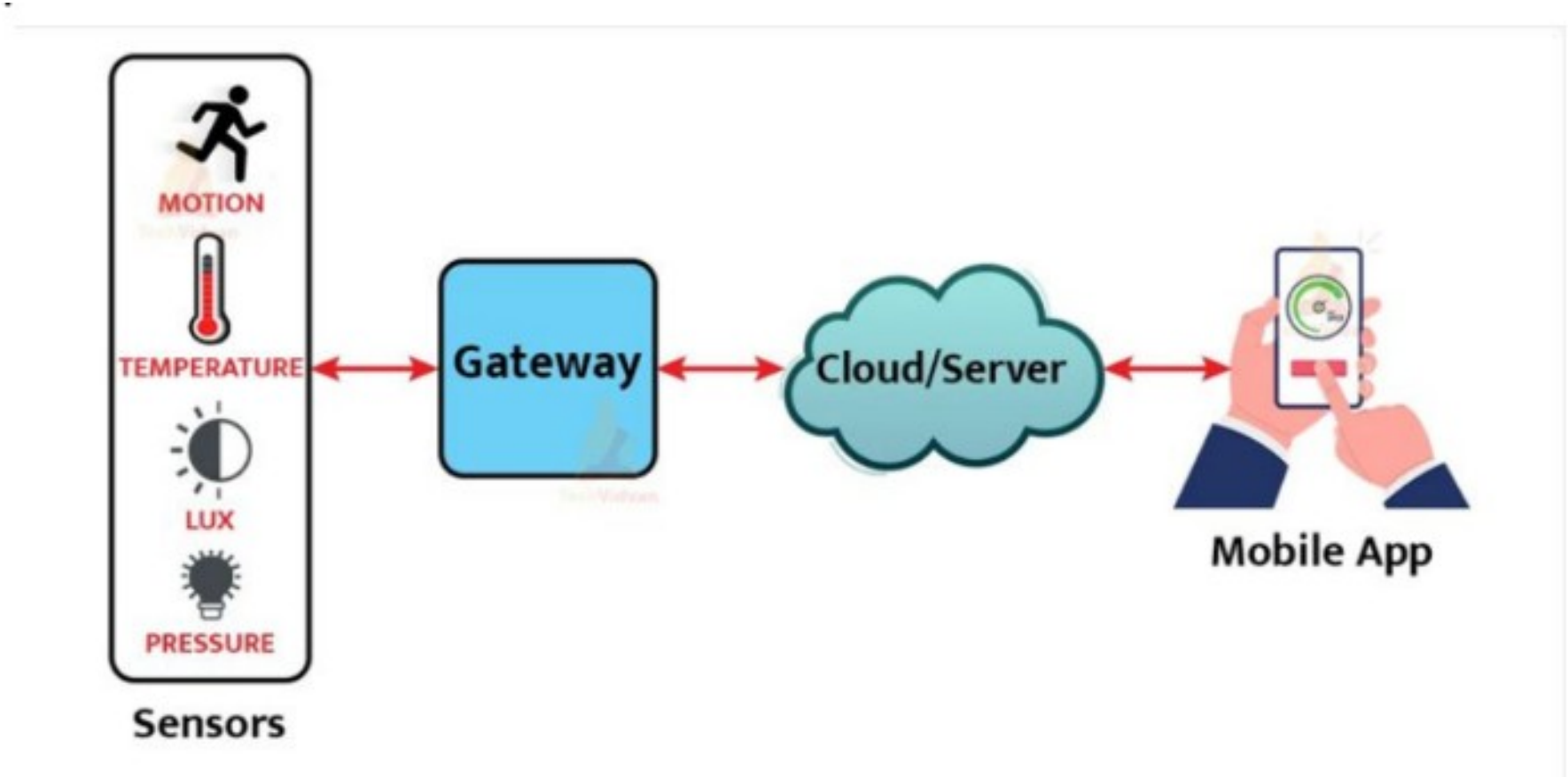
CLOUD stores, processes and analyses the data

4

USER INTERFACE shows insight and sends commands back

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How IoT Works



Sensors and devices talk to the cloud; software processes the data and acts, or shows it to a user.

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What Is a Sensor?

1

Definition

A device that detects an event or change in its environment and turns it into a signal.

2

Temperature

Thermocouples, thermistors, RTDs and semiconductor sensors — the most measured quantity.

3

Humidity & pressure

Environmental monitoring for comfort, process control and weather sensing.

4

Proximity & motion

Detect the presence, absence or movement of an object without contact.

5

Gas & chemical

Air quality, CO_2 and hazardous-gas detection in cities and industrial plants.

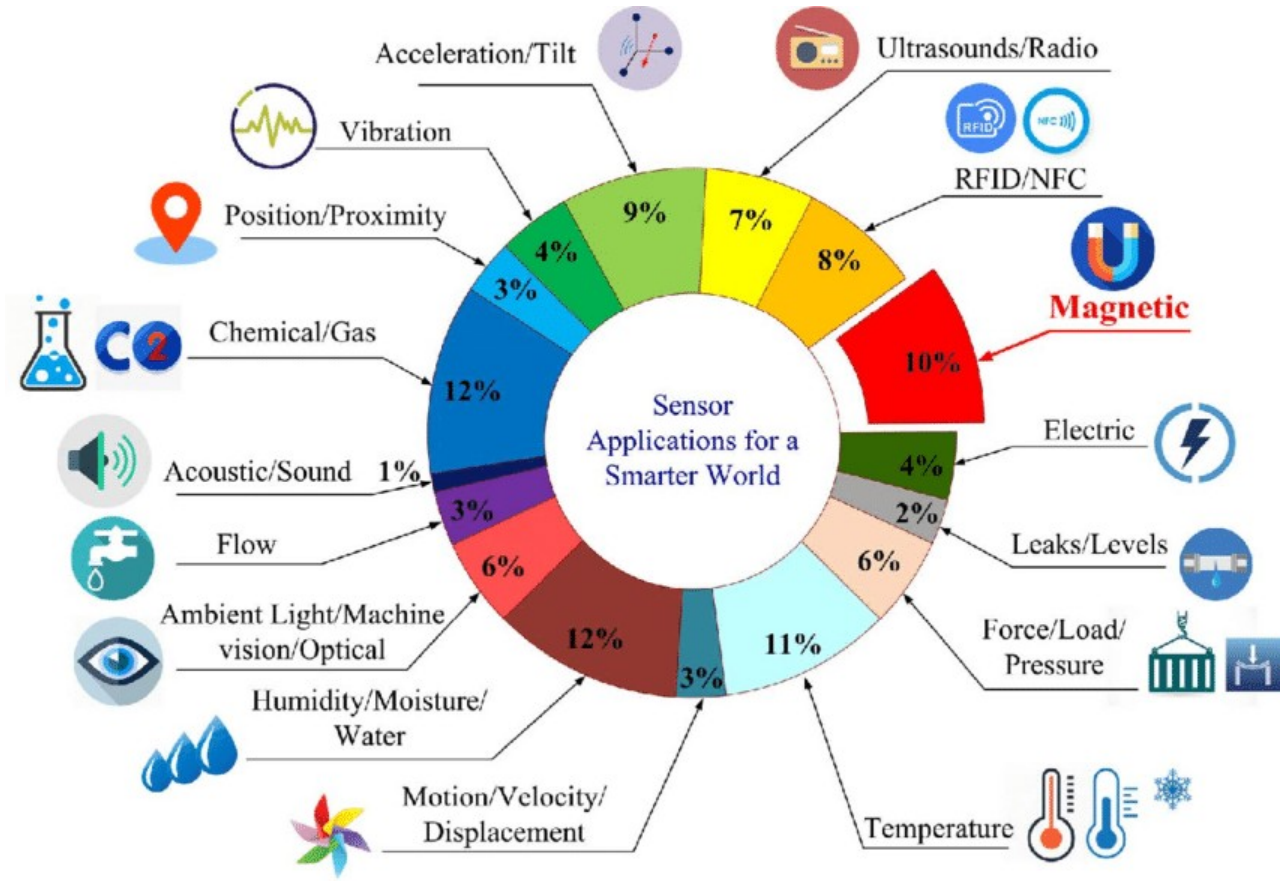
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Light & IR

Ambient light sensing, obstacle detection and remote-control technology.

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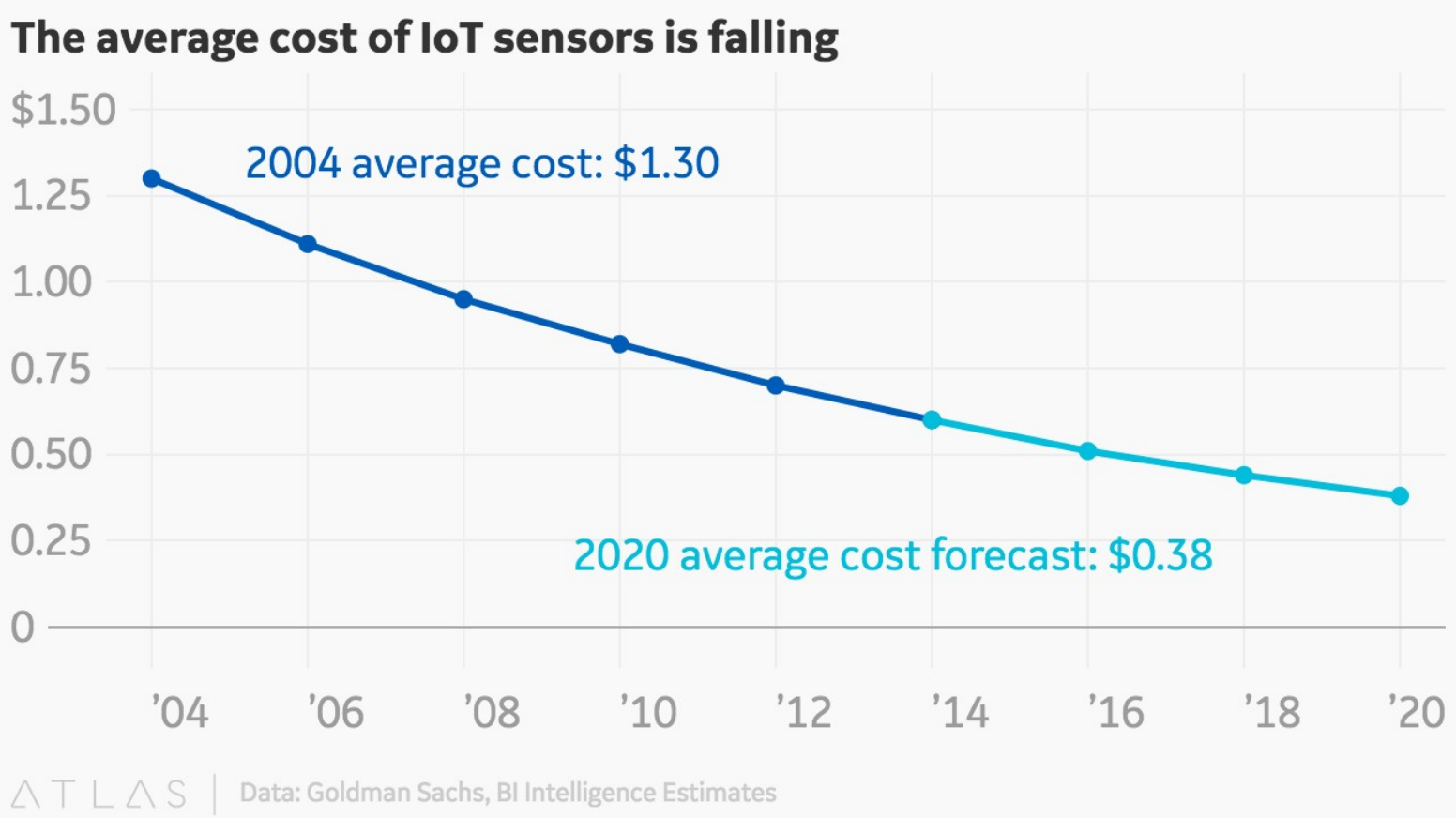
Sensors for IoT



Low-cost sensors are what made large-scale IoT deployment economically viable.

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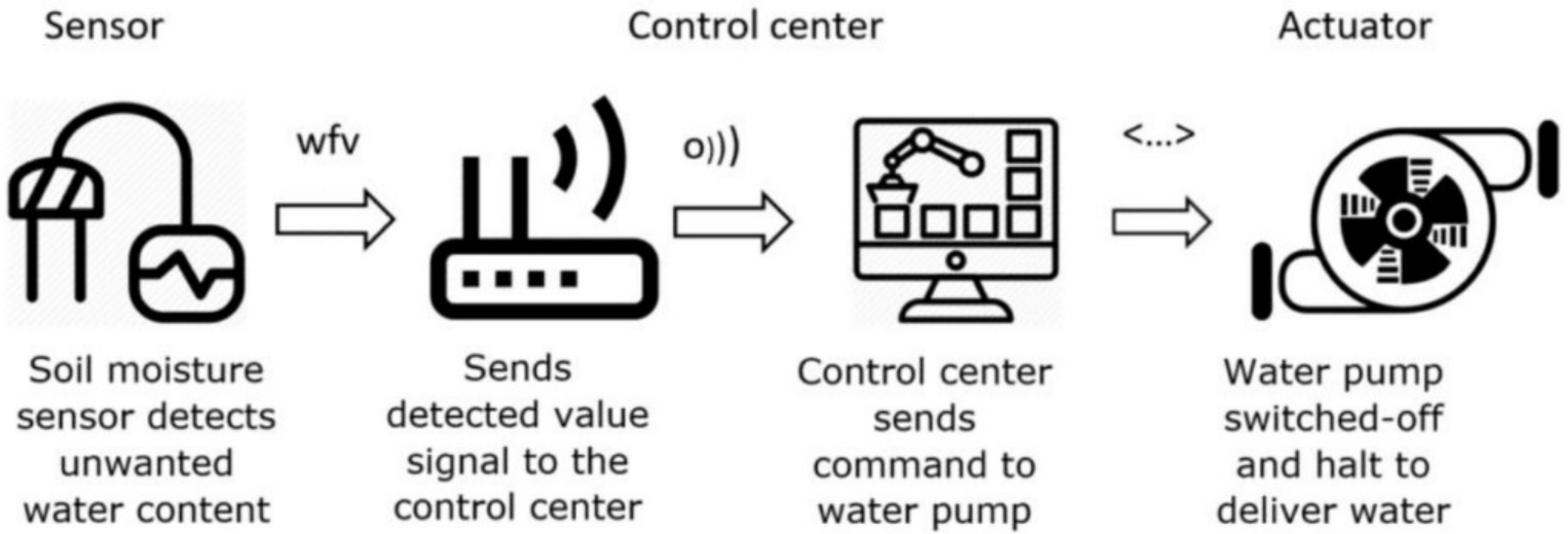
More Sensors for IoT



Every physical quantity a business cares about has a low-cost sensor for it.

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Actuators for IoT



Actuators turn a decision in the cloud back into a physical action.

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Sensors, Actuators and Gateways

Sensors — sense

- Temperature & humidity
- Motion, presence, vibration
- Light, gas, air quality
- GPS location, current

Actuators — act

- Relays and switches
- Motors and servos
- Valves and pumps
- Alarms and displays

Gateways — connect

- Aggregate nearby devices
- Translate protocols
- Buffer when offline
- Add edge processing

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CONNECTIVITY CHOICES

Wireless Technologies for IoT

1

WiFi

High bandwidth, short range, mains power — best indoors where power is available.

2

Bluetooth / BLE

Very low power, very short range — wearables and personal devices.

3

Zigbee / Z-Wave

Low-power mesh — smart buildings and home automation.

4

Cellular (4G/5G)

Wide coverage with a subscription cost — vehicles, logistics and remote assets.

5

LoRaWAN / Sigfox

Long range, very low power, tiny payloads — agriculture, metering, city-scale sensing.

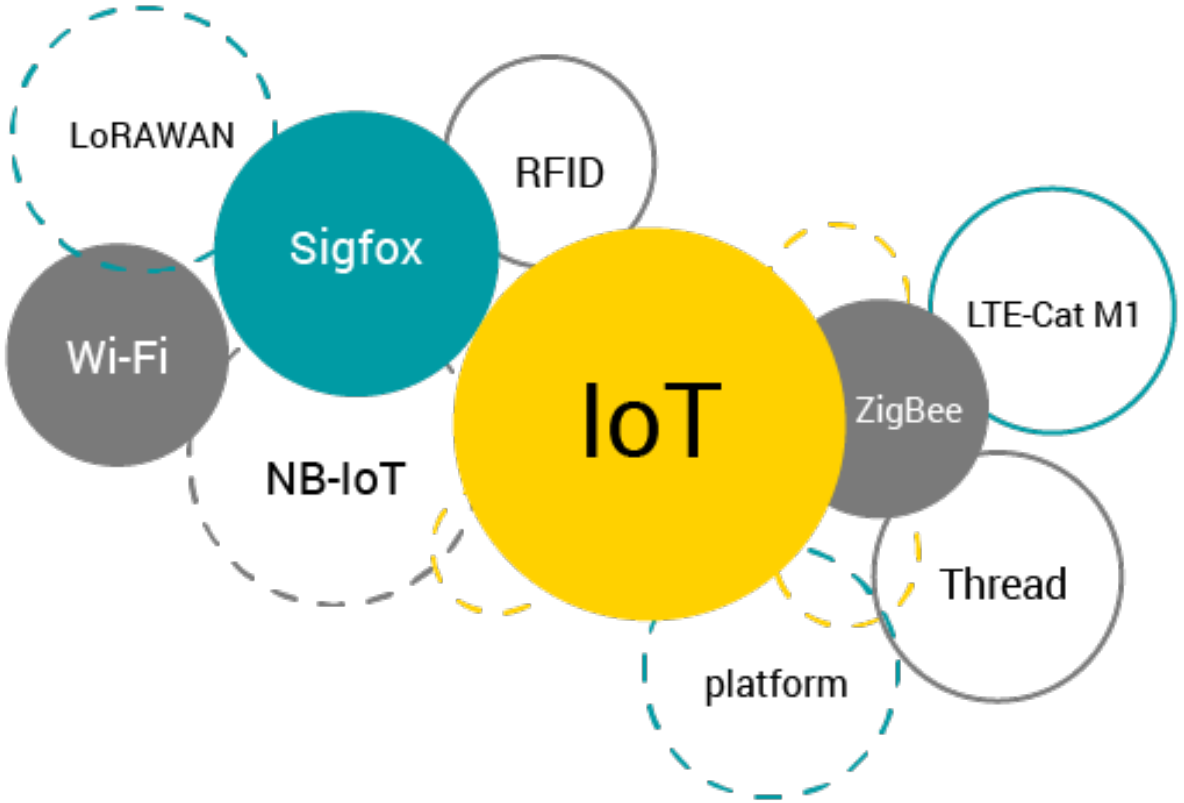
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NB-IoT

Licensed low-power wide-area networking with carrier-grade reliability.

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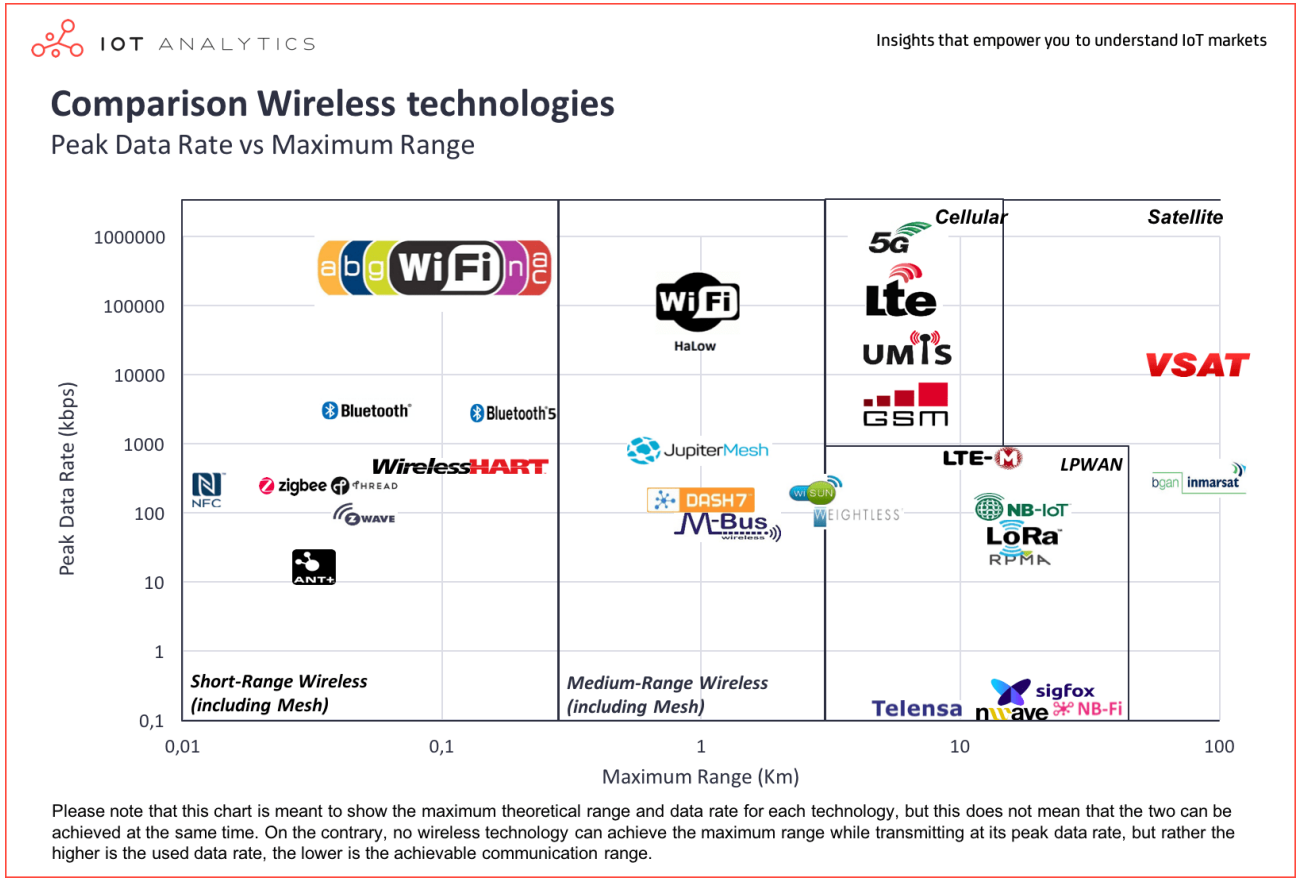
Wireless Technologies for IoT



Range, power consumption, bandwidth and cost determine which technology fits a use case.

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Comparing Wireless Technologies



There is no single best option — every choice trades range against power, bandwidth and cost.

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FOUR CATEGORIES

Types of IoT

1

Consumer IoT

Everyday use — home appliances, voice assistants, smart lighting and wearables.

2

Commercial IoT

Healthcare and transport — smart pacemakers, monitoring systems, connected cars.

3

Industrial IoT (IIoT)

Manufacturing and energy — digital control systems and industrial big data.

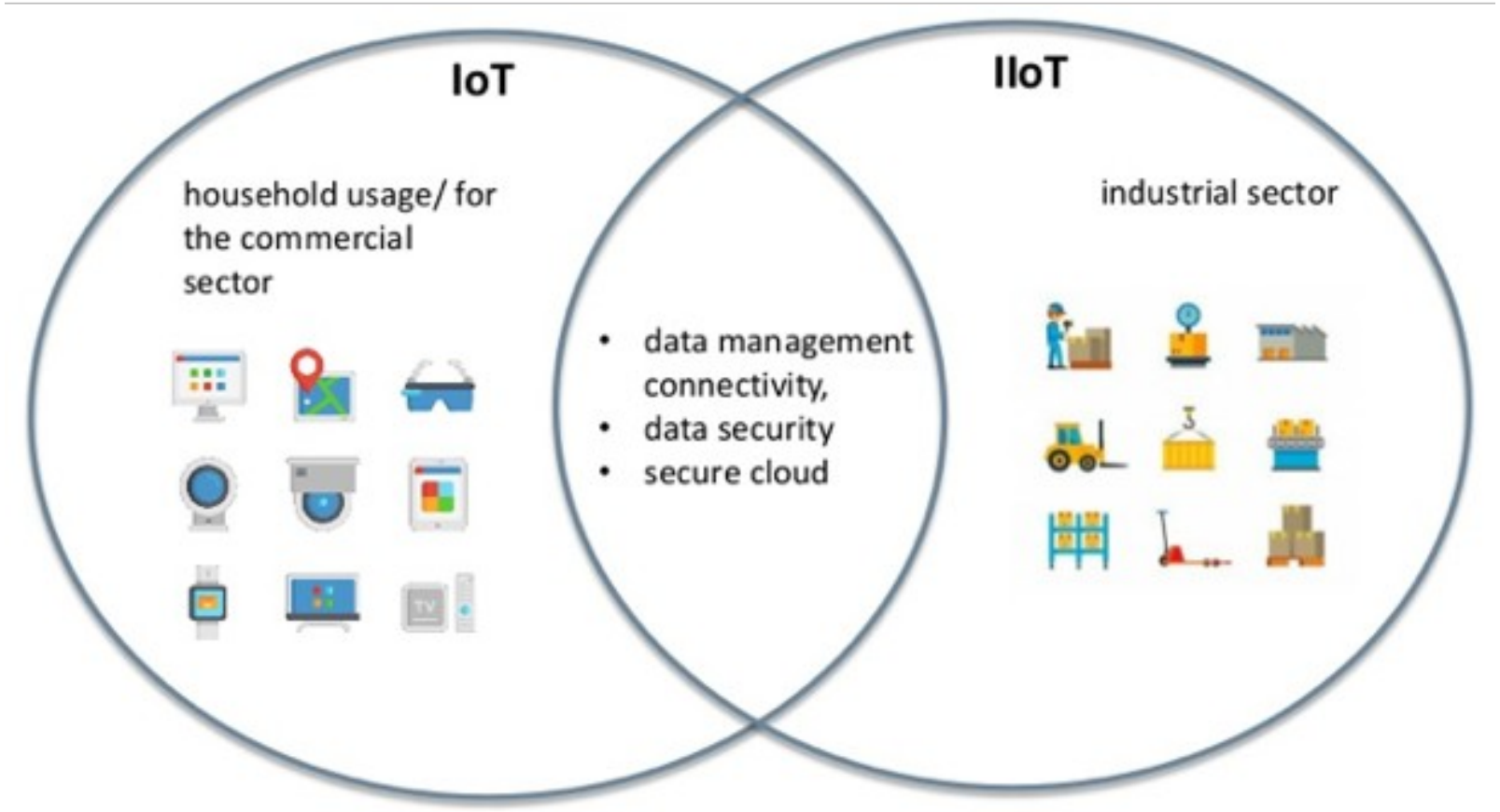
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Infrastructure IoT

Smart cities — infrastructure sensors and city management systems.

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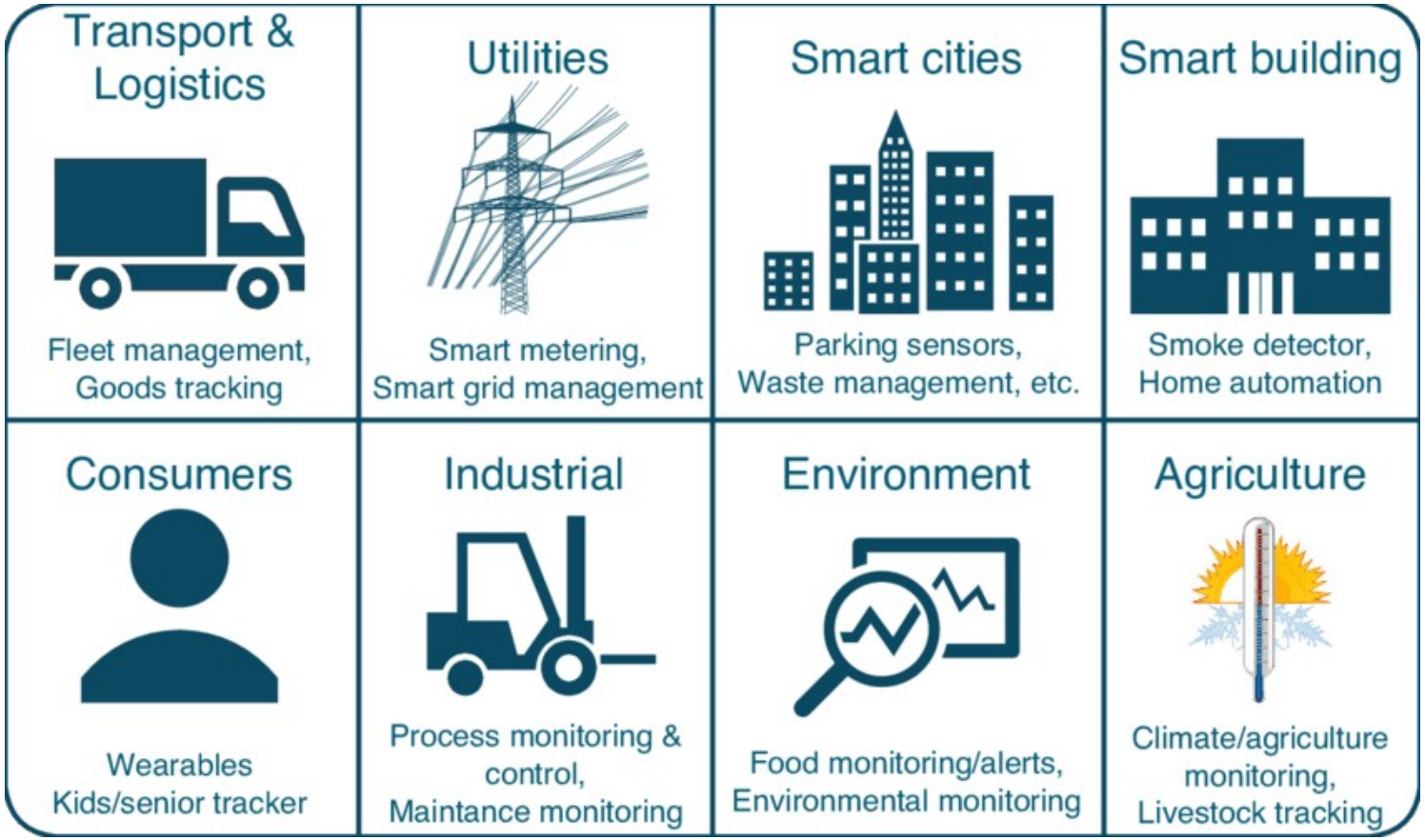
IoT vs Industrial IoT (IIoT)



IIoT extends IoT into industrial sectors, where reliability and safety requirements are higher.

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IoT Applications Across Industry



IoT has found its way into every conceivable field, from agriculture to aerospace.

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Use Case — Healthcare



Asset tracking and patient monitoring: find the nearest wheelchair, watch vitals continuously.

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Use Case — Agriculture



Soil, weather and crop sensing drive precision irrigation and fertiliser decisions.

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Use Case — Smart Home



Smart outlets, thermostats, locks and leak detectors improve efficiency and safety.

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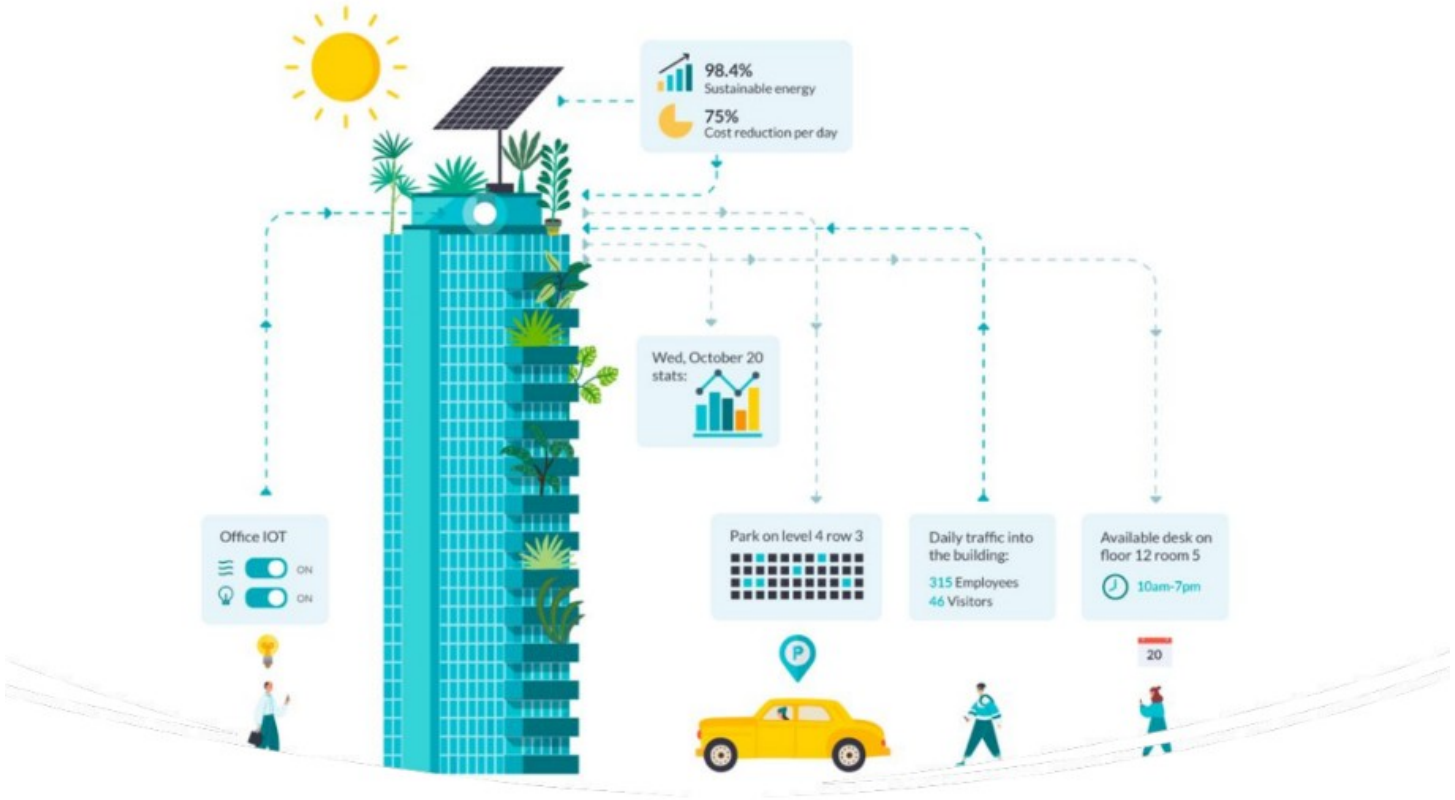
Use Case — Smart City



Air quality, smart lighting, infrastructure monitoring and parking management.

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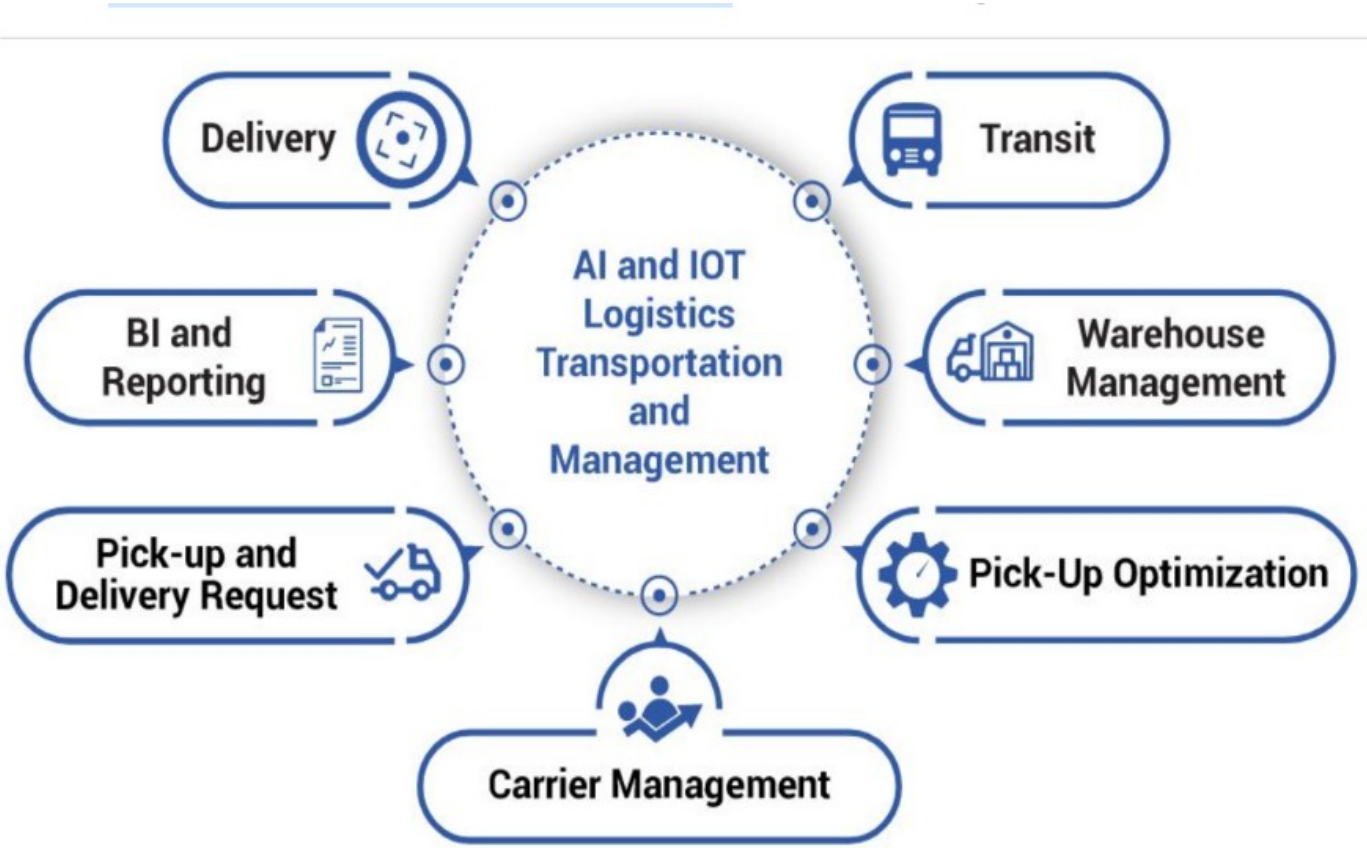
Use Case — Smart Building



Occupancy, energy and equipment sensing cut utility cost and enable predictive maintenance.

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Use Case — Logistics



Track-and-trace with temperature control — critical for food, flowers and pharmaceuticals.

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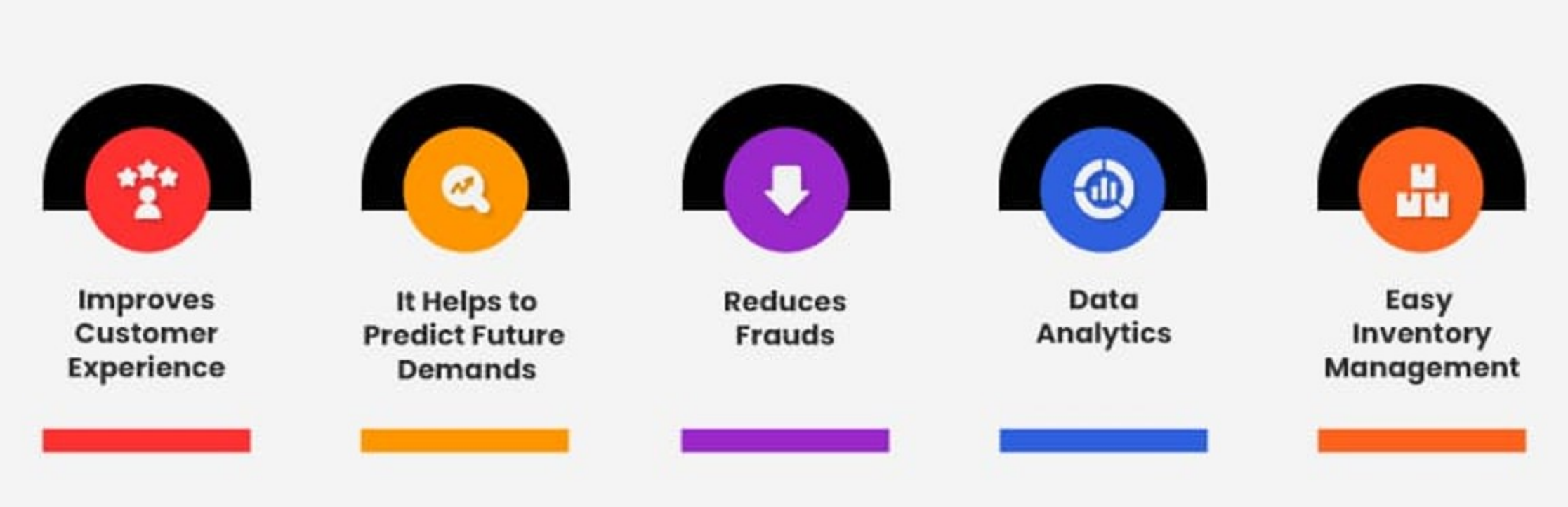
Use Case — Manufacturing



Production-line monitoring enables proactive maintenance before equipment fails.

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Use Case — Retail



Smart shelves and beacons manage inventory and personalise the customer experience.

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TOPIC 01 · ACTIVITY

Discussion — Analyse an IoT Use Case

DISCUSSION · 20 MIN

In pairs, pick one of the IoT use cases covered (healthcare, agriculture, connected vehicles, smart home, smart city, smart building, logistics, manufacturing or retail) and analyse how the data collected leads to a recommended control action.

1. What is physically being sensed, how often, and by what kind of sensor?
2. Which wireless technology suits that deployment — and why does range or power decide it?
3. What control action does the data recommend, and who or what carries it out?
4. What would the organisation lose if that measurement did not exist?

YOUR OUTPUT A short analysis linking the sensed data to a specific recommended control action.

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TOPIC RECAP

Recap — Overview of Internet of Things (IoT)

- IoT connects physical objects to the internet so they can collect and exchange data.
- Every IoT system has four blocks: sensors/devices, connectivity, cloud processing and a user interface.
- Sensors turn a physical phenomenon into a signal; actuators turn a decision back into physical action.
- Wireless choice trades range against power, bandwidth and cost — there is no single best option.

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Lunch Break

1 hour · Digital attendance (PM) on your return

02

TOPIC 02

Collect and Post Environmental Data to Cloud

Cloud platforms · Raspberry Pi · Sensors · Device tokens · HTTP REST & MQTT uplink

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LO2 · A2, K2, K3 · Day 1 PM

Key Concepts — Collect and Post Environmental Data to Cloud

1

Cloud computing for IoT

The cloud gives constrained devices the storage, compute and availability they lack.

2

IoTFlow

Our platform: projects, devices, tokens, dashboards, alerts and automations in one place.

3

Raspberry Pi as gateway

A low-cost Linux computer with GPIO, WiFi and Python — it reads sensors and speaks to the cloud.

4

Reading environmental data

Temperature, humidity and pressure from a Sense HAT, DHT11/DHT22 or a simulated source.

5

Device tokens

A token authenticates every message a device sends — and is why credential management matters.

6

HTTP REST & MQTT uplink

POST JSON to a REST endpoint, or publish over MQTT for lightweight real-time streaming.

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Cloud Computing for IoT

1

Why the cloud

Constrained devices lack the storage, compute and availability the data needs.

2

Always available

The platform keeps receiving data whether or not your laptop is switched on.

3

Scales with the fleet

One device or a thousand — the same endpoint and the same code.

4

Data persistence

History is retained so trends and performance can be analysed later.

5

Accessible anywhere

A browser or a phone shows the same live data from any location.

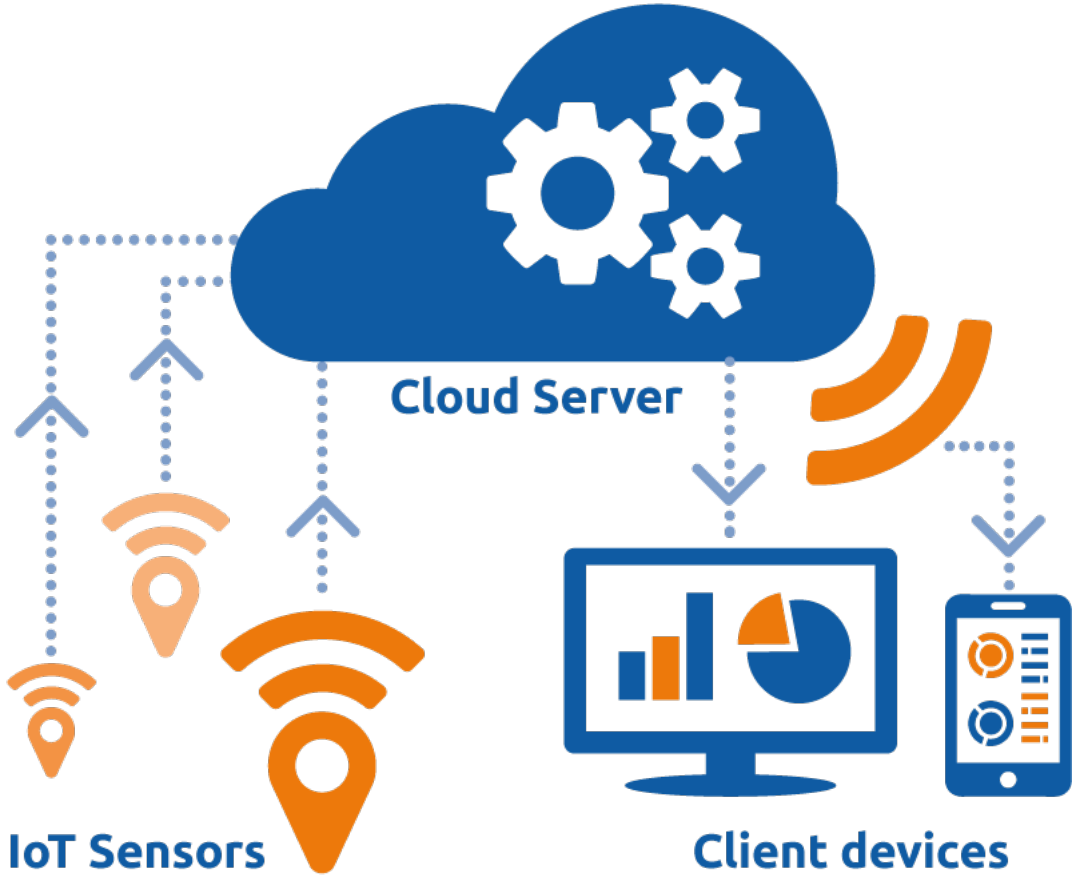
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Integrations

Webhooks and APIs push events into the systems the business already uses.

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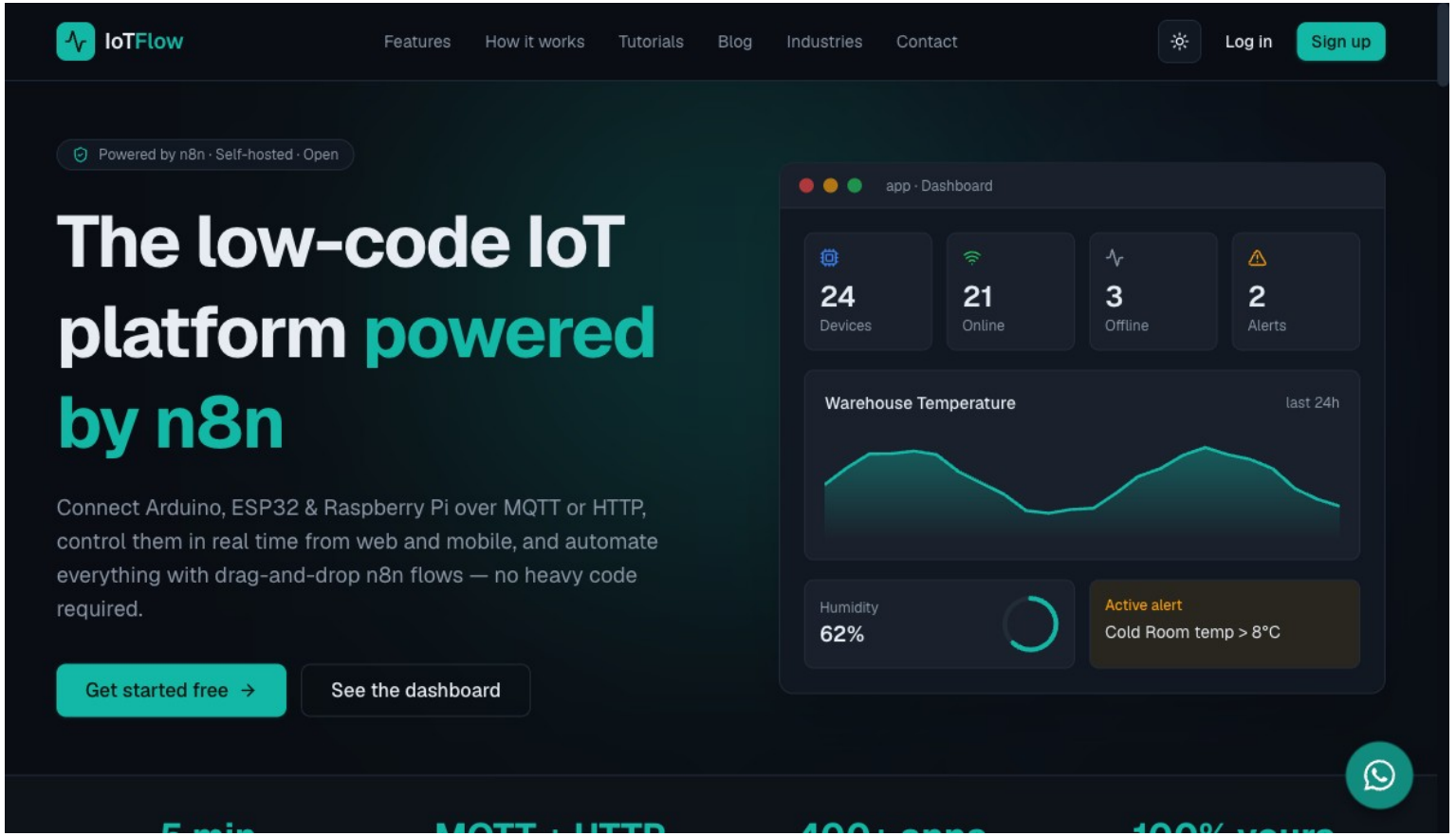
Cloud Computing for IoT



The cloud gives constrained devices the storage, compute and availability they lack.

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IoTFlow — https://iot.tertiaryinfotech.com



The platform you will use for every lab in this course.

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https://iot.tertiaryinfotech.com

IoTFlow — Our IoT Platform

1

Projects

A self-contained workspace with its own devices, dashboard, alerts, map and automations.

2

Devices & tokens

Register a device in a guided wizard; a device token authenticates every message it sends.

3

HTTP & MQTT

POST JSON to a REST endpoint, or publish/subscribe over MQTT for real-time two-way traffic.

4

Dashboards

Number cards, gauges, charts, LED indicators and maps that refresh automatically.

5

Alerts & automations

Threshold and offline rules that fire a webhook into an n8n workflow.

6

Clients

Official Python client (pip install iotflow) and an Arduino library for ESP32/ESP8266.

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The Raspberry Pi as an IoT Gateway

1

A full Linux computer

Low cost and credit-card sized, running Raspberry Pi OS with Python built in.

2

GPIO pins

Connect sensors and actuators directly to the board — no extra hardware needed.

3

Built-in WiFi

Reaches the cloud without a separate network module.

4

Python ecosystem

sense_hat, gpiozero, adafruit-dht and the iotflow client are a pip install away.

5

Runs unattended

systemd restarts your sensor script automatically after a power cut.

6

Remote access

SSH or VNC lets you work on the Pi headless, from your laptop.

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THREE WAYS TO GET A READING

Reading Environmental Data

Sense HAT

- Temperature, humidity, pressure
- Motion and orientation
- 8×8 LED matrix and joystick
- One import: sense_hat

DHT11 / DHT22

- Temperature and humidity
- Connects to a single GPIO pin
- Very low cost
- adafruit_dht library

Simulated data

- No hardware required
- Random or replayed values
- Useful for testing the uplink
- Identical platform behaviour

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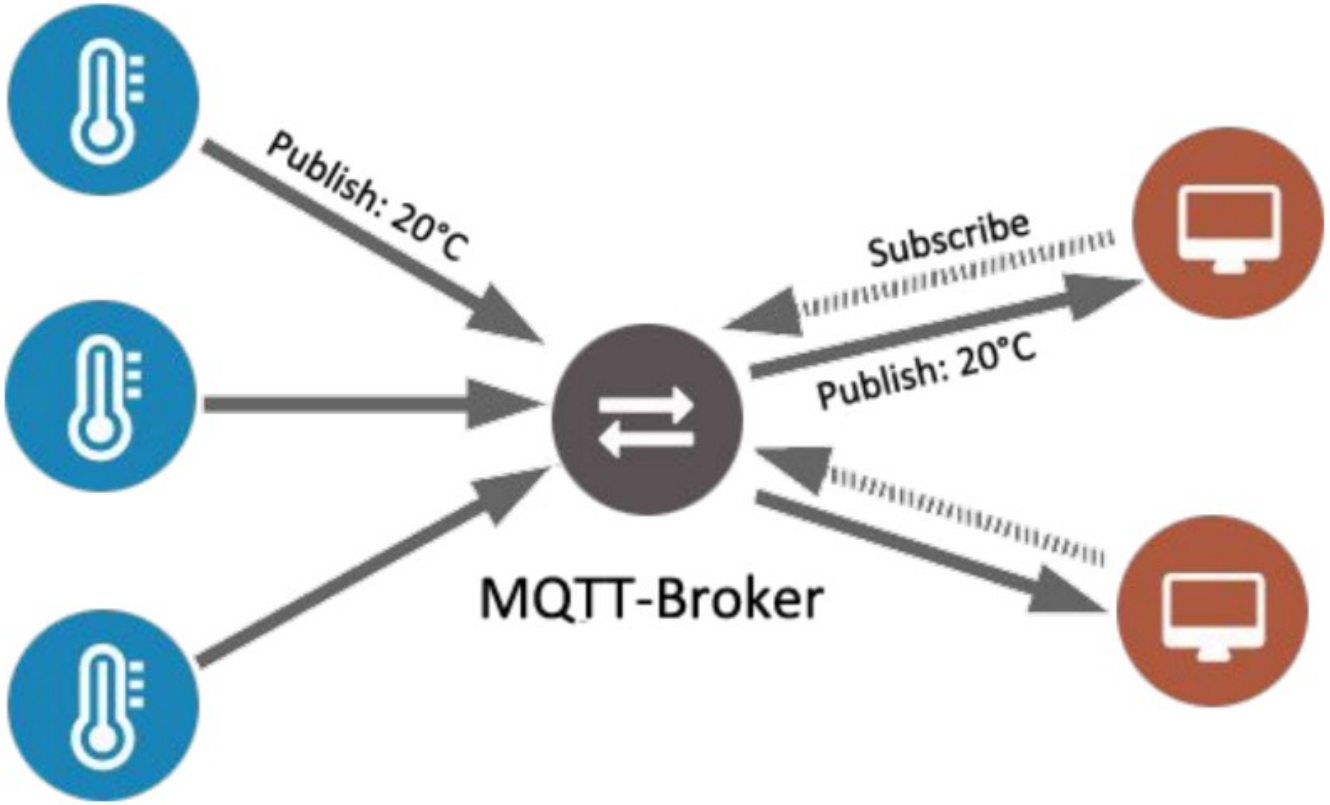
FROM SENSOR TO ACTION AND BACK

The IoT Data Round Trip

1. DEVICE reads a sensor value
2. UPLINK posts it over HTTP or MQTT
3. CLOUD stores, charts and evaluates rules
4. DECISION — a person or an automation acts
5. DOWNLINK sends a command back to the device

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How MQTT Works



Publish/subscribe: devices publish to a topic and the broker delivers to every subscriber.

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HTTP REST vs MQTT

Simple

- HTTP REST
  - Request/response
  - Simplest to start with
  - POST JSON to an endpoint
  - Works from cURL or any language
  - Higher overhead per message

Real-time

- MQTT
  - Publish/subscribe via a broker
  - Lightweight and low bandwidth
  - Persistent connection
  - Best for real-time two-way control
  - Ideal for many devices

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Device Tokens and Security

1

What a token is

A secret string that authenticates every message the device sends.

2

Shown once

Copy it when the device is created — the platform will not display it again.

3

Anyone holding it can post

A leaked token means someone else can publish as your device.

4

Keep it out of code

Store it in an environment variable, never in a Git repository.

5

Use TLS endpoints

Encrypt the data in transit so the token cannot be read off the wire.

6

Rotate if exposed

Regenerate the token immediately if you suspect it has leaked.

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TOPIC 02 · PRACTICAL

Hands-On Practical — Labs 1 and 2

Running on IoTFlow (https://iot.tertiaryinfotech.com). Step-by-step instructions are in your Learner Guide — follow along there.

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TOPIC 02 · HANDS-ON

Register Your Raspberry Pi on IoTFlow

LAB 1

Identify an area for an IoT solution and connect the asset to the cloud (A2, K2).

Turn your Raspberry Pi into a managed IoT device. You sign in to IoTFlow, create a project workspace for your monitoring scenario, register the Pi through the Add Device wizard, choose how it will talk to the cloud (HTTP or MQTT), and capture the device token that authenticates every message it sends.

YOU'LL BUILD

A registered Raspberry Pi device in your own IoTFlow project, with its device token saved

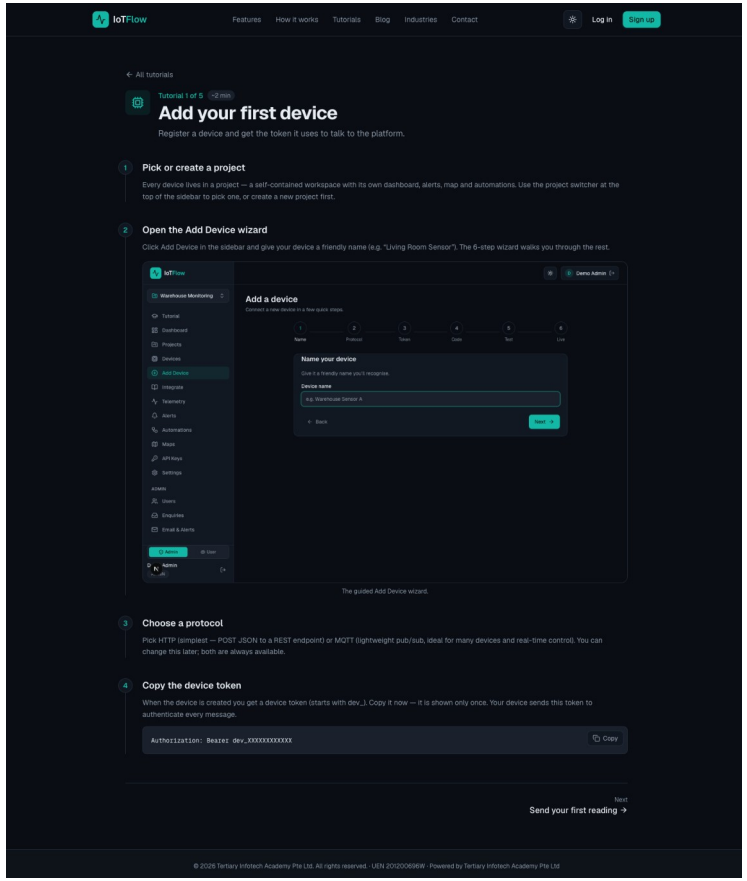
WHY THE BUSINESS CARES

This is the first step of any IoT deployment: an asset that was previously invisible now has an identity, an owner and a data stream that can be measured, monitored and managed.

TOOLS IoTFlow projects · Add Device wizard · Device tokens · HTTP & MQTT

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Lab 1 — Register Your Raspberry Pi on IoTFlow



IoTFlow — A registered Raspberry Pi device in your own IoTFlow project, with its device token saved

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TOPIC 02 · HANDS-ON

Collect Environmental Data and Post It to the Cloud

LAB 2

Collect environment data with the Raspberry Pi and post it to cloud computing (A2, K2, K3).

Read temperature and humidity on the Raspberry Pi and stream the readings to IoTFlow — first with a single cURL call, then with the official Python client over HTTP, then continuously over MQTT. This is the uplink every IoT solution depends on: without a reliable data stream there is nothing to analyse.

YOU'LL BUILD

Live temperature and humidity telemetry streaming from the Raspberry Pi into IoTFlow

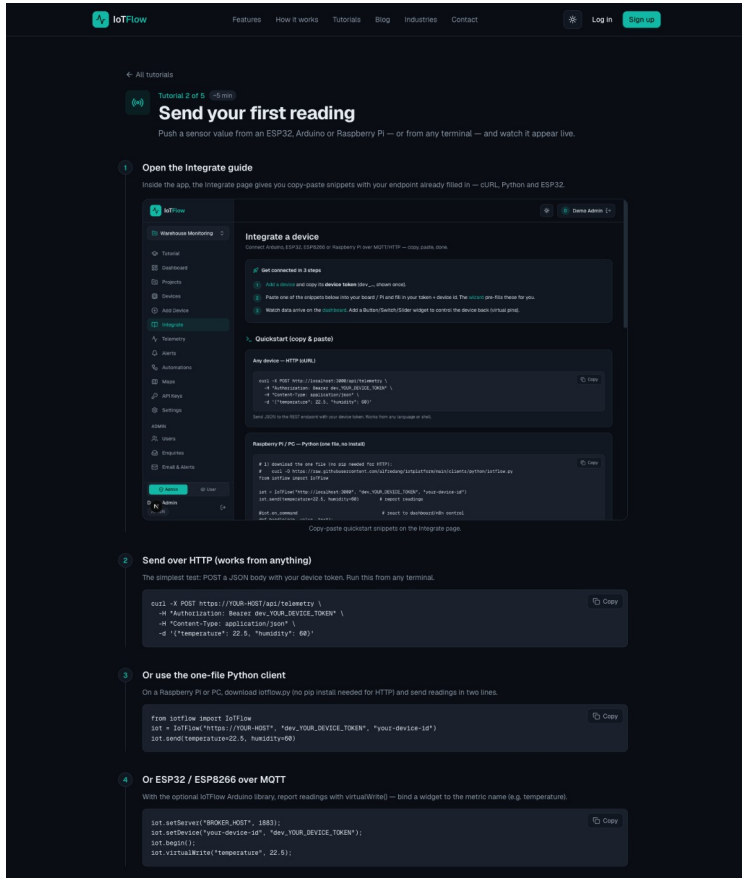
WHY THE BUSINESS CARES

Each reading is an operational fact with a timestamp. Once the stream exists, the organisation can answer questions it previously could not — how often, how long, how bad, and at what cost.

TOOLS Sense HAT / DHT22 · Telemetry REST API · Python client (pip install iotflow) · MQTT broker

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Lab 2 — Collect Environmental Data and Post It to the Cloud



IoTFlow — Live temperature and humidity telemetry streaming from the Raspberry Pi into IoTFlow

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TOPIC RECAP

Recap — Collect and Post Environmental Data to Cloud

- The cloud gives a constrained device the storage, availability and reach it does not have on its own.
- The Raspberry Pi reads a sensor over GPIO and posts the reading to the cloud over HTTP or MQTT.
- A device token authenticates every message — treat it like a password.
- Once the uplink is reliable, the business finally has data it can act on.

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SEE YOU AT 9:30AM

End of Day 1

Tomorrow: turn the data you are now collecting into a dashboard, an alert, and an automatic control action.

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2

DAY 2

Analytics, Visualisation and Control — From Data to Action

Topic 3: IoT Data Analytics and Visualization · Topic 4: Monitor Data to Trigger Control from Cloud

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03

TOPIC 03

IoT Data Analytics and Visualization

Dashboards · Number cards, gauges & charts · Maps · Alert rules · Analysing trends

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WHY VISUALISE

Raw readings are not insight.

A number in a database changes nothing. A dashboard that makes a problem obvious — and an alert that arrives in time — is what changes the outcome.

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LO3 · A3, K4, K5 · Day 2 AM

Key Concepts — IoT Data Analytics and Visualization

1

Why visualise

A dashboard turns a stream of raw readings into an operational picture a manager can act on.

2

Display widgets

Number cards, gauges, line and bar charts, LED indicators and maps, refreshed automatically.

3

Analysing the data

Compare current readings against history to spot trends, drift and abnormal behaviour.

4

Alert rules

Threshold and device-offline rules fire the moment a reading crosses a business-meaningful limit.

5

Recording performance

Telemetry history is the evidence base for measuring whether the solution actually works.

6

From data to decision

Analytics only pays off when a reading changes what somebody — or something — does next.

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Choosing the Right Widget

1

Number card

One current value at a glance — the reading somebody asks for most often.

2

Gauge

A value inside its safe operating range — instantly shows 'how close to the limit'.

3

Line chart

History over time — the only way to see trends, drift and daily cycles.

4

Bar chart

Comparing totals across periods or across devices.

5

LED indicator

A binary state: on/off, open/closed, healthy/faulty.

6

Map

Where the device is — essential for vehicles, logistics and field assets.

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From Reading to Decision

1. COLLECT telemetry from the device
2. VISUALISE it on a dashboard
3. ANALYSE the history for trends and drift
4. DETECT when a threshold is crossed
5. DECIDE and record the action taken

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Analysing IoT Data

1

Current vs normal

Compare the live reading against the established operating range.

2

Trends over time

A slow drift upward is often more important than a single high reading.

3

Peaks and troughs

The extremes reveal whether the process ever leaves its safe envelope.

4

Noise vs event

One odd sample is noise; a sustained change is a real event worth acting on.

5

Correlation

Temperature rising as humidity falls tells a story neither metric tells alone.

6

Recording performance

The telemetry history is the evidence that the solution actually works.

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Alert Rules That Work

1

Threshold rules

Fire when a metric crosses a limit: temperature > 30, humidity < 40.

2

Device-offline rules

Fire when a device stops reporting — silent failure is the biggest risk.

3

Set a meaningful limit

Choose the value at which a person would genuinely need to act.

4

Avoid alert fatigue

Too many alerts and the real one gets ignored — tune the threshold.

5

Active and resolved

Track the alert lifecycle so response time can be measured.

6

Trigger automation

An alert can call a webhook and start a workflow with no human involved.

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TOPIC 03 · PRACTICAL

Hands-On Practical — Lab 3

Running on IoTFlow (https://iot.tertiaryinfotech.com). Step-by-step instructions are in your Learner Guide — follow along there.

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TOPIC 03 · HANDS-ON

Analyse and Visualise Your IoT Data on the Cloud

LAB 3

Use the platform's tools to analyse IoT data and record performance (A3, K4, K5).

Build a live dashboard from the readings your Raspberry Pi is streaming — number cards, gauges, line charts, LED indicators and a map — then interrogate the history to spot trends and set a threshold alert at the point where somebody would actually need to act.

YOU'LL BUILD

A live monitoring dashboard for your Raspberry Pi, plus a working threshold alert rule

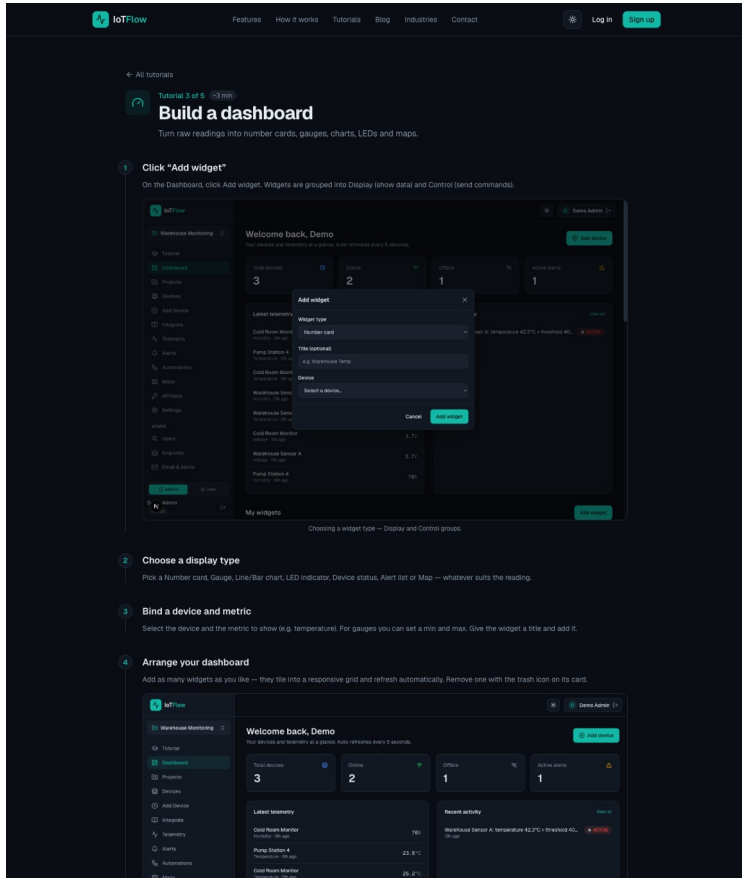
WHY THE BUSINESS CARES

This is the screen the supervisor actually looks at. A dashboard makes performance visible, and an alert makes a problem timely instead of discovered too late.

TOOLS Display widgets · Number cards · Gauges · Line charts · Maps · Alert rules · Telemetry history

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Lab 3 — Analyse and Visualise Your IoT Data on the Cloud



IoTFlow — A live monitoring dashboard for your Raspberry Pi, plus a working threshold alert rule

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Discussion — Design Your Monitoring Dashboard

DISCUSSION · 20 MIN

Working from the data your Raspberry Pi is now streaming, decide what a supervisor actually needs to see on one screen, and which threshold would justify interrupting somebody.

1. Which three widgets earn their place on the screen, and what does each one answer?
2. What is the normal operating range, and at what value does it become a problem?
3. How would you tell a genuine fault apart from a single noisy reading?
4. What performance record would you need to prove the solution is working?

YOUR OUTPUT A dashboard layout plus one justified alert threshold with its rationale.

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TOPIC RECAP

Recap — IoT Data Analytics and Visualization

- A dashboard turns a stream of readings into an operational picture somebody can act on.
- Pick the widget that answers the question: number for now, gauge for range, chart for trend.
- Analyse history, not just the latest value — trends and drift matter more than one sample.
- An alert rule set at a business-meaningful threshold turns monitoring into timely action.

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Lunch Break

1 hour · Digital attendance (PM) on your return

04

TOPIC 04

Monitor Data to Trigger Control from Cloud

Downlink control · Virtual pins · Automations & n8n · Monitoring effectiveness · IoT security

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LO4 · A4, K6, K7 · Day 2 PM

Key Concepts — Monitor Data to Trigger Control from Cloud

1

Reading data back

Query the current state of every metric and virtual pin over the REST API or MQTT.

2

Downlink control

Virtual pins carry commands from the dashboard back to the device to switch a relay or set a value.

3

Closing the loop

Monitoring tells you what happened; control lets the system act without anyone driving to site.

4

Automation with n8n

Alerts fire a webhook into a low-code workflow that notifies, logs, and can ask AI to recommend.

5

Monitoring effectiveness

Track uptime, alert response and the metric you set out to improve — then tune the rules.

6

IoT security

Constrained, unattended devices with physical consequences — protect tokens, transport and firmware.

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TWO DIRECTIONS, TWO PURPOSES

Uplink vs Downlink

Monitoring

- Uplink — telemetry
  - Device → cloud
  - Sensor readings and status
  - HTTP POST /api/telemetry
  - or MQTT publish
  - This is your monitoring data

Control

- Downlink — commands
  - Cloud → device
  - Switch a relay, set a value
  - MQTT topic devices/<id>/down
  - or HTTP state polling
  - This is your automation

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Control Widgets and Virtual Pins

1

Virtual pin

A named channel (V1, relay, pump) carrying a command from dashboard to device.

2

Switch

A toggle for a persistent on/off state — a relay, a fan, a pump.

3

Button

A momentary action — reset, trigger, open for two seconds.

4

Slider

A value within a range — motor speed, brightness, a setpoint.

5

Terminal

Free-text commands for diagnostics and advanced control.

6

@on_command

The handler on the Pi that receives the value and drives the GPIO pin.

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Closing the Monitor-and-Control Loop

1. SENSE — the Pi reads the temperature
2. REPORT — telemetry arrives in the cloud
3. DETECT — the alert rule crosses its threshold
4. AUTOMATE — n8n notifies, logs and decides
5. ACT — a downlink command switches the fan on

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Automation with n8n

1

Webhook trigger

IoTFlow calls an n8n webhook when your chosen event fires.

2

Notify

Email, Telegram or WhatsApp so the right person hears about it immediately.

3

Record

Append the event to Google Sheets or a database to build the performance history.

4

Analyse with AI

An AI node summarises recent readings and recommends an action in plain language.

5

Control

An HTTP Request node sends a downlink command straight back to the device.

6

No code

The whole workflow is drag-and-drop — no server and no deployment.

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TOPIC 04 · PRACTICAL

Hands-On Practical — Lab 4

Running on IoTFlow (https://iot.tertiaryinfotech.com). Step-by-step instructions are in your Learner Guide — follow along there.

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TOPIC 04 · HANDS-ON

Monitor Data and Trigger Control on the Raspberry Pi

LAB 4

Monitor IoT data and trigger a control action on the device from the cloud (A4, K6, K7).

Read device state back out of the cloud over the REST API and MQTT, then add Control widgets (switch, slider) bound to virtual pins and drive an LED or relay on the Raspberry Pi from the dashboard. Reading data is monitoring; sending a command back is automation — and that is where the labour saving appears.

YOU'LL BUILD

A dashboard switch and slider that control an LED/relay on the Raspberry Pi in real time

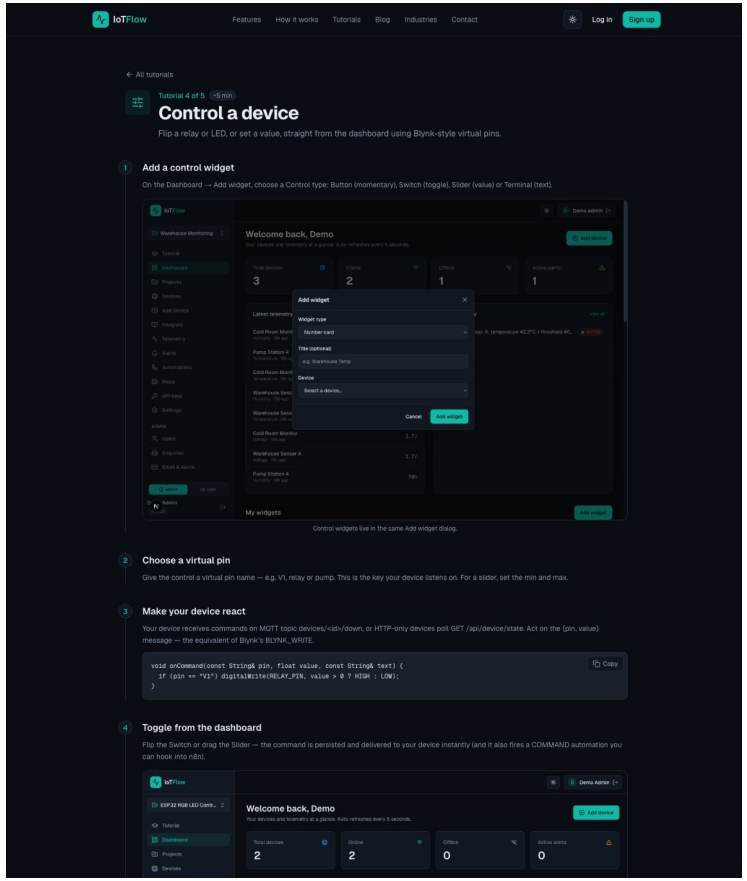
WHY THE BUSINESS CARES

Monitoring tells you what happened. Control lets the system act without a person driving out to site — usually the single largest line in an IoT cost-benefit case.

TOOLS REST API (GET /api/device/state) · MQTT subscribe · Control widgets · Virtual pins · @on_command · GPIO

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Lab 4 — Monitor Data and Trigger Control on the Raspberry Pi



IoTFlow — A dashboard switch and slider that control an LED/relay on the Raspberry Pi in real time

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Monitoring the Effectiveness of an IoT Solution

1. **Device uptime**
   What percentage of the time is the device actually reporting?

2. **Data completeness**
   Are readings arriving at the expected interval, with no silent gaps?

3. **Alert response time**
   How long from the threshold being crossed to the action being taken?

4. **False-positive rate**
   How many alerts turned out not to need any action at all?

5. **The business metric**
   Did downtime, energy, spoilage or labour actually improve?

6. **Tune and repeat**
   Adjust thresholds and sampling rates based on what the record shows.

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IoT Security — Why It Is Different

Constrained devices

- Microcontrollers, not servers
- Often battery powered
- May not support TLS/SSL

Unattended & long-lived

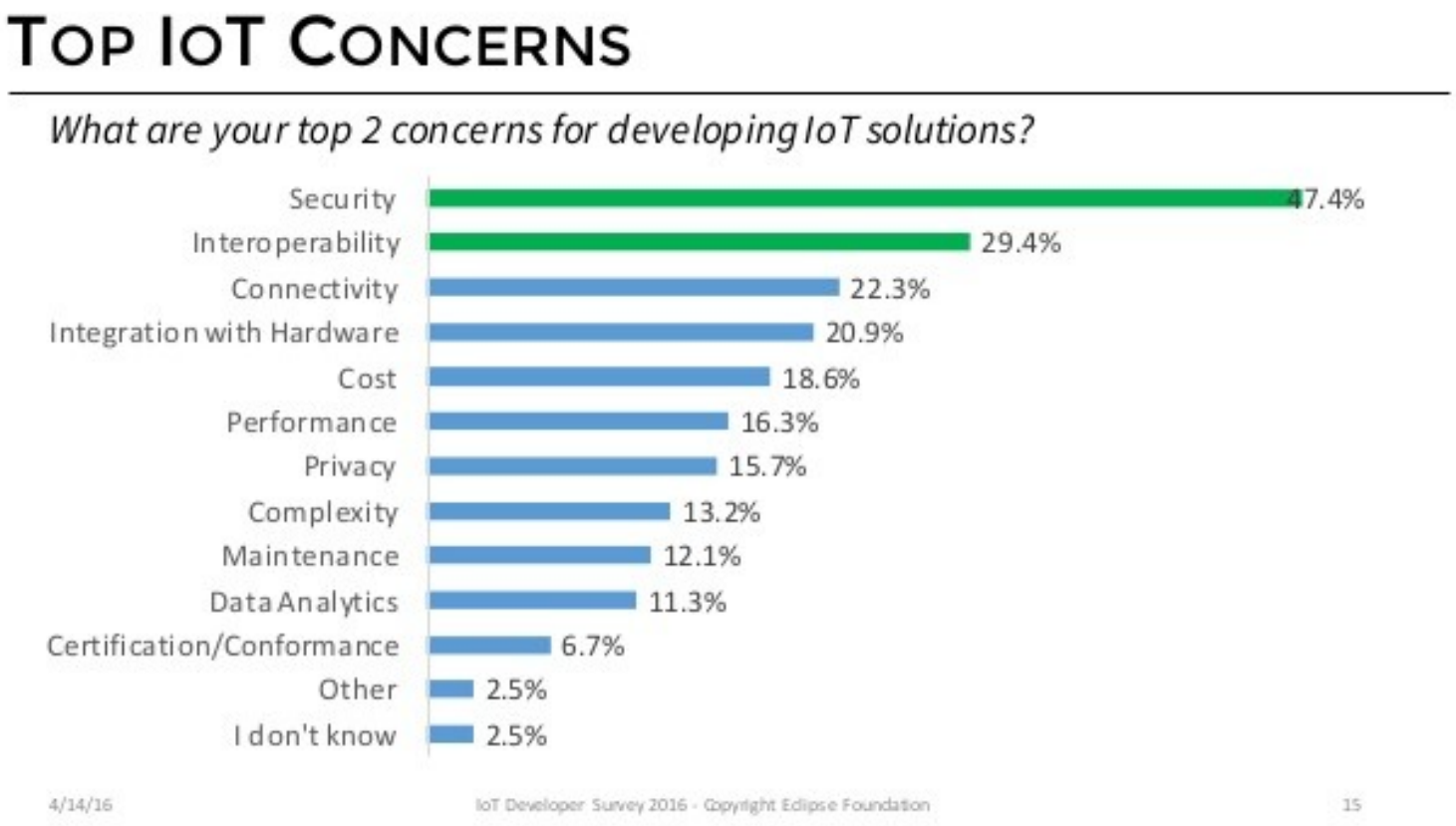
- Run for years without supervision
- Patching is inconvenient
- Rogue devices go unnoticed

Physical consequences

- IT breach leaks data
- IoT breach can cause harm
- Safety-critical systems at risk

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IoT Security Is a Top Concern



Security and data privacy are consistently the biggest barriers to IoT adoption.

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CASE STUDY · WHY DEVICE SECURITY MATTERS

The Mirai botnet, 12 October 2016.

Default credentials on cameras and routers were harvested into a botnet whose DDoS attack took much of the US east coast offline — an unsecured device is everyone's problem.

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Securing Your IoT Solution

1

Change default credentials

The single most common cause of IoT compromise — Mirai proved it.

2

Protect the device token

Environment variables, never source control; rotate it if exposed.

3

Encrypt in transit

Use TLS endpoints so readings and tokens cannot be read off the wire.

4

Keep firmware updated

Unpatched, unattended devices are the easiest target on any network.

5

Segment the network

Keep IoT devices off the network carrying business-critical systems.

6

Least privilege

A device should be able to do its job and nothing more.

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IoT Guidelines and Communication Standards

1

MQTT

OASIS-standard lightweight publish/subscribe messaging — the IoT default.

2

HTTP / REST

Universally supported request-response, easiest to integrate and debug.

3

CoAP

A compact request-response protocol designed for very constrained devices.

4

IEEE 802.15.4

The physical/MAC layer beneath Zigbee and other low-rate wireless networks.

5

LoRaWAN

LoRa Alliance specification for long-range, low-power wide-area networking.

6

IMDA & PDPA

Singapore's IoT cyber-security guide and personal-data obligations.

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TOPIC 04 · ACTIVITY

Discussion — Monitor Effectiveness and Manage Risk

DISCUSSION · 20 MIN

Take the IoT solution you have built across the labs and plan how you would monitor its effectiveness in production — and how you would secure it.

1. Which metrics tell you the solution is working — uptime, response time, or the business outcome?
2. How would you know a device had silently stopped reporting?
3. Where are the credentials in your solution, and what happens if one leaks?
4. Which standard or guideline would you cite when asked how the data is protected?

YOUR OUTPUT A monitoring and security plan: the effectiveness metrics, the offline strategy, and the top three risks.

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TOPIC RECAP

Recap — Monitor Data to Trigger Control from Cloud

- Reading data back and sending commands forward turns monitoring into automation.
- Virtual pins carry the command from the dashboard to a handler that drives the GPIO pin.
- Alerts plus n8n notify, record, analyse with AI and can act with no human in the loop.
- Measure effectiveness — uptime, response time and the business metric — and secure every layer.

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WRAP-UP

Course Summary & Next Steps

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LEARNING OUTCOMES

What You Achieved

1

LO1 · Analysed IoT technologies

Sensors, actuators, wireless technologies and use cases across nine industries.

2

LO2 · Collected and posted data

Registered a Raspberry Pi and streamed live sensor readings over HTTP and MQTT.

3

LO3 · Analysed and visualised

Built a live dashboard, analysed telemetry history and set a working alert rule.

4

LO4 · Monitored and controlled

Drove hardware from the cloud and automated the response end to end.

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AFTER TODAY

Your Next Steps

1

Keep your IoTFlow project

Your devices, dashboard and automations stay available after the course.

2

Deploy your Pi for real

Pick one thing worth measuring and let it run unattended for a week.

3

Revisit the Learner Guide

Every lab's step-by-step instructions are there for you to repeat.

4

Explore the tutorials

More platform walkthroughs at https://iot.tertiaryinfotech.com/tutorials

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CONTINUE YOUR LEARNING

Recommended Courses

- WSQ - Internet of Things (IoT) Fundamental for Beginners
- WSQ - Business Innovation with Internet of Things (IoT)
- WSQ - Practical Electronics Design with Arduino Microcontroller
- WSQ - Hands-On Guide to IoT Development with Microcontrollers
- Browse the full catalogue at www.tertiarycourses.com.sg

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Summary & Q&A

- IoT connects physical assets so an organisation can measure what used to be invisible.
- The Raspberry Pi reads the physical world; the cloud stores, analyses and acts on it.
- Dashboards make performance visible, alerts make problems timely, automation closes the loop.
- Any questions before the assessment?

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ASSESSMENT · 120 MINUTES TOTAL

Final Assessment

WA (Q&A) — 60 min

- Short-answer questions
- Assesses knowledge K1–K7
- Open book

Practical Performance — 60 min

- Build a working IoT solution
- Assesses abilities A1–A4
- Open book, done individually

Before you begin

- Take the Assessment digital attendance
- Complete the TRAQOM survey
- Slides + Learner Guide allowed
- Submit as directed by your assessor

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ON ASSESSMENT DAY

Assessment Flow

1. TRAQOM survey — scan the QR code on the LMS
2. Assessment digital attendance — scan the SSG QR
3. Sit the WA (Q&A) — 60 min, open book
4. Complete the Practical Performance — 60 min
5. Submit your answers on the LMS
6. Sign the Assessment Summary Record

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Digital Attendance & TRAQOM Survey (Mandatory)

1

Assessment attendance

Take the Assessment digital attendance before you begin — scan the SSG QR code.

2

Cert & TRAQOM survey

Complete the mandatory survey at https://lms-tms.tertiaryinfotech.com/

3

Your certificate

Your certificate is issued through the LMS once you are assessed Competent.

4

75% minimum

At least 75% attendance is required to be eligible for assessment and funding.

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WE'RE HERE TO HELP — DURING AND AFTER THE CLASS

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SEE YOU AT THE NEXT COURSE

Thank You!

You can now build, analyse and control a complete IoT solution with a Raspberry Pi and the cloud.

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